

STAGE-1 RESEARCH REPORT

Explainable Hybrid Deep Learning-Based Intrusion Detection System for IoT Networks

A Critical Literature Synthesis, State-of-the-Art Analysis, and Proposed Research Methodology

Minor Project — Stage 1 Report

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Abstract

Internet of Things (IoT) networks are increasingly targeted by botnet, denial-of-service, reconnaissance, spoofing, and injection attacks, while remaining constrained in computation, memory, and update capability. This Stage-1 report critically reviews the literature (2018–August 2026) on IoT intrusion detection, tracing the progression from signature- and anomaly-based detection through classical machine learning to deep learning, hybrid deep learning (CNN–LSTM, CNN–GRU, attention-enhanced recurrent networks, autoencoder–classifier pipelines, Transformers, and Graph Neural Networks), and post-hoc explainable AI (SHAP, LIME, Integrated Gradients). The synthesis shows that near-perfect benchmark accuracy is common but frequently unreliable because of dataset leakage, random row-level splitting, and class imbalance, and that explanation methods are rarely evaluated for fidelity, stability, or robustness rather than merely being produced as illustrative plots. Public IoT intrusion datasets, including N-BalIoT, IoT-23, Bot-IoT, TON_IoT, Edge-IIoTset, IoTID20, and CIIoT2023, are compared for scale, realism, and known weaknesses. Because the combination of hybrid deep learning with SHAP/LIME explainability has already been explored extensively, this report argues that a genuinely defensible undergraduate contribution lies not in inventing a new architecture but in a leakage-aware, generalization-tested, and explanation-quality-evaluated study. Accordingly, a compact CNN–GRU architecture evaluated on Edge-IIoTset (with an optional CIIoT2023 cross-dataset test) under random, chronological, and group-based validation, paired with SHAP-based explanations subjected to fidelity and stability testing, is proposed as the Stage-2 project direction, together with a complete experimental methodology, baseline set, ablation plan, evaluation metrics, limitations, and future scope.

1. Introduction

The Internet of Things (IoT) refers to networks of devices that sense or act upon the physical world and communicate through wired or wireless network interfaces. The U.S. National Institute of Standards and Technology (NIST) characterizes an IoT device as a system possessing at least one transducer and at least one network interface, using technologies such as Ethernet, Wi-Fi, Bluetooth, LTE, and Zigbee.

1.1 IoT Security Context

IoT deployments span perception/device, network/transport, edge/fog, cloud/service, and application layers, each associated with distinct security concerns ranging from weak credentials and insecure firmware at the device layer to injection, exfiltration, and unauthorized control at the application layer. NIST identifies device identification, secure configuration, data protection, logical access control, secure software update, cybersecurity state awareness, and device security as core cybersecurity capabilities that IoT devices are expected to support.

IoT security is difficult in practice because devices are often deployed for long periods with infrequent firmware updates, use default credentials, expose management interfaces, and lack sufficient CPU, memory, storage, or battery capacity to run conventional endpoint-security agents.

1.2 IoT Attack Surface

The attack surface of an IoT deployment includes physical device interfaces, firmware and OS vulnerabilities, weak or default credentials, exposed Telnet/SSH/HTTP/MQTT/CoAP services, wireless links, cloud APIs and mobile applications, gateways, supply-chain components, inter-device trust relationships, and data pipelines. Lightweight IoT protocols such as MQTT and CoAP are frequently deployed with weak or absent authentication and encryption, which increases reliance on behavior-based network monitoring where device-level controls are incomplete.

1.3 Why IoT Intrusion Detection Is Different

Conventional enterprise intrusion detection research typically assumes relatively powerful and homogeneous hosts, stable infrastructure, and centralized monitoring. IoT environments violate these assumptions: device populations are heterogeneous in hardware, OS, protocol, and traffic pattern; new devices join frequently; a single gateway may serve thousands of devices; devices may be physically inaccessible after deployment; privacy constraints can prevent centralizing raw traffic; and compromised devices are commonly recruited into botnets. As a consequence, a model that performs well on one laboratory dataset may fail when deployed against a different device population or a new attack campaign — a central motivation for the generalization-focused methodology proposed later in this report.

2. Intrusion Detection Systems: Fundamentals and Taxonomy

2.1 Definition

NIST defines an intrusion detection system as hardware or software that automates the process of monitoring system or network events and analyzing them for evidence of security problems (NIST SP 800-94). IDS may be classified by what they monitor (network, host, wireless, application) and by how they detect malicious behaviour (signature-based, anomaly-based, specification-based, or hybrid).

2.2 IDS Classification by Monitoring Target

Type	Monitoring target	Typical evidence
NIDS	Network traffic	Flow statistics, packets, ports, protocols, TCP flags
HIDS	Individual host/device	Logs, processes, system calls, files, configuration changes
Wireless IDS	Wireless communication	Association behaviour, radio frames, wireless anomalies
Network-behaviour IDS	Network-wide behaviour	Traffic volumes, connection graphs, communication patterns
Application IDS	Application activity	API calls, web requests, command patterns

2.3 Detection Strategies

Signature-based detection

Signature-based IDS compares observed events against a database of known attack patterns. It offers high precision on known attacks and is comparatively interpretable, but is weak against zero-day attacks and previously unseen IoT botnet variants, and requires continuous signature maintenance.

Anomaly-based detection

Anomaly-based IDS models normal behaviour and flags deviations, making it central to nearly all IoT ML/DL literature because it can, in principle, detect unknown attacks. Its main weaknesses are a high false-positive risk, difficulty in defining 'normal' in dynamic IoT settings, and vulnerability to concept drift.

Specification-based detection

Specification-based IDS relies on manually engineered rules describing permitted device behaviour (for example, that a temperature sensor may contact only a designated broker). It can be interpretable and effective in tightly controlled environments but requires substantial device-specific engineering.

Hybrid IDS

The term 'hybrid' is used inconsistently across the literature. It may denote an architectural combination (e.g., CNN + LSTM), an algorithmic combination (ML + DL, supervised + unsupervised), a deployment-level combination (edge + cloud), a privacy-level combination (federated + centralized), or a data-level combination (flows + logs + telemetry). A rigorous report should therefore state explicitly which sense of 'hybrid' is being used rather than treating the term as self-evident — a discipline followed throughout this report and adopted for the proposed project.

2.4 Centralized, Edge, Fog, Cloud, and Federated IDS

Architecture	Main processing location	Advantages	Main limitations
Centralized/cloud IDS	Raw data sent to central server	High computational capacity, global visibility	Latency, bandwidth, privacy, single point of failure
Edge IDS	Gateway or local edge server	Lower latency, less bandwidth, local privacy	Limited CPU, memory and storage
Fog IDS	Intermediate distributed nodes	Hierarchical scalability, aggregation	More complex orchestration
Hybrid edge–cloud IDS	Fast filtering at edge, deep analysis in cloud	Balances latency and capacity	Synchronization and deployment complexity
Federated IDS	Local training, shared model updates	Raw data remains local	Non-IID data, communication cost, poisoning risk

For an undergraduate prototype, a centralized gateway simulation is the most practical starting point; edge-oriented characteristics can be approximated by measuring latency, memory, model size, and CPU usage on commodity hardware, while a full federated deployment is more realistically treated as future work.

3. IoT Attack Taxonomy

A working knowledge of representative IoT attack classes is required to interpret dataset labels and to design meaningful detection experiments. Table 3.1 summarizes major categories that recur across the surveyed literature and datasets.

Table 3.1 — Representative IoT Attack Categories

Attack	High-level operation	Typical detection difficulty
DoS	Single source overwhelms a service/device	Usually easier when volumetric
DDoS	Many compromised sources coordinate a DoS	Harder — individual sources may look moderate
Mirai / botnet	Scans for default-credential devices, recruits them	Easier for known patterns; harder for variants
Brute force	Repeated login attempts	Detectable via authentication metadata
Reconnaissance/scanning	Discovers hosts, ports, services, vulnerabilities	Can resemble legitimate discovery traffic
Spoofing	Falsifies source/destination/DNS/ARP identity	Difficult with flow-only data
MITM	Intercepts/modifies communications	Requires topology or protocol-level evidence
Injection (command/SQL/XSS)	Inserts malicious payloads into services	Flow-only models may miss payload semantics
Malware / ransomware	Establishes persistence, control, or extortion	Novel variants may evade signature matching
Data exfiltration	Unauthorized extraction of data	Legitimate cloud sync creates ambiguity

The CICIoT2023 dataset alone contributes 33 distinct attacks generated against a topology of 105 IoT devices and grouped into DDoS, DoS, reconnaissance, web-based, brute-force, spoofing, and Mirai categories, while Edge-IIoTset covers 14 attacks across DoS/DDoS, information gathering, MITM, injection, and malware categories, and N-BaIoT is restricted to Mirai and BASHLITE infections across nine commercial devices. This diversity of taxonomies is itself a source of difficulty when comparing results across studies, since attack definitions and groupings are not standardized between datasets.

4. Machine Learning for IoT Intrusion Detection

4.1 Traditional Supervised Models

Because most IoT IDS datasets consist of tabular flow records rather than raw packet sequences, classical models remain strong and necessary baselines. Tree ensembles in particular are important: a proposed deep model should be compared against Random Forest and XGBoost, not only against weak neural baselines.

Model	Strengths	Weaknesses	IoT suitability
Logistic Regression	Fast, simple, interpretable	Linear decision boundary	Excellent gateway baseline
Decision Tree	Easy to explain, handles nonlinear rules	Prone to overfitting	Good interpretable IDS
Random Forest	Strong tabular performance, robust	Larger model; importance can be biased	Very suitable for flow-based IDS
SVM	Effective in high dimensions, small data	Expensive at scale	Suitable for smaller subsets

Model	Strengths	Weaknesses	IoT suitability
XGBoost	Excellent tabular performance	Training/tuning cost	Strong gateway model and XAI baseline
Isolation Forest / One-Class SVM	Unsupervised anomaly detection	Sensitive to contamination	Useful for unknown-attack experiments

4.2 Learning Paradigms

Supervised learning, using labelled benign/attack records, usually produces the strongest benchmark performance but cannot reliably detect attack families absent from training and is sensitive to label quality and dataset shortcuts. Unsupervised learning (autoencoders, clustering, Isolation Forest, one-class methods) is useful when labels are incomplete and can detect deviations from normal behaviour, but typically exhibits a higher false-positive rate and requires a clean or mostly benign training set. Semi-supervised learning, which combines abundant unlabeled traffic with limited labelled attacks, is realistic for IoT but adds implementation complexity, while self-supervised learning (masked-attribute prediction, temporal reconstruction, contrastive learning) is a promising but non-essential direction for a Minor Project.

5. Deep Learning for IoT Intrusion Detection

5.1 DNN / MLP

A multilayer perceptron treats each flow as a feature vector. It is straightforward to implement and computationally moderate, but does not naturally model time or network topology and is generally less interpretable than tree models. A 2023 IoT IDS study using a four-layer fully connected network across blackhole, DDoS, opportunistic-service, sinkhole, and wormhole attacks reported an average accuracy of 93.74%, illustrating that MLP baselines remain a reasonable — though non-temporal — reference point.

5.2 CNN and 1D-CNN

A 1D-CNN applies convolutional filters over ordered features or short sequences, capturing local feature interactions efficiently and in a manner well suited to edge deployment. Its main weakness is that the ordering of tabular features may be artificial, and high results can arise from leakage if inputs contain IP, port, or scenario identifiers. A 2025 study in Cybersecurity proposed an explainable CNN approach using SHAP to identify important network-flow features, illustrating the now-common pattern of a CNN classifier paired with post-hoc explanation.

5.3 RNN, LSTM, and GRU

Recurrent architectures are suited to temporally ordered flows. LSTM uses gated memory cells to preserve long-term dependencies and is strong for temporal attack patterns but is more computationally expensive than GRU. GRU uses fewer gates, yielding a lower parameter count, faster training, and often a favourable trade-off for edge-oriented prototypes, at some cost to expressiveness on very long dependencies.

Bidirectional variants (BiLSTM/BiGRU) can improve offline classification but are not always appropriate for real-time streaming detection, since future observations are unavailable at decision time.

5.4 CNN–LSTM and CNN–GRU

In these hybrids, a CNN extracts local patterns and a recurrent layer models temporal relationships (flow sequence → CNN feature extraction → LSTM/GRU temporal modeling → classifier). This combination is heavily researched: recent studies apply CNN–LSTM, CNN–GRU, LSTM–CNN, and CNN–LSTM–attention models to IoT, IIoT, and general NIDS datasets. Its principal conceptual limitation is that many datasets consist of independently generated flow rows with no genuine temporal sequence; if rows are arbitrarily reshaped into sequences, the recurrent layer may learn row-ordering artifacts rather than authentic network dynamics — a caution directly relevant to how any hybrid model in this project should be justified and evaluated.

5.5 Attention-Enhanced Hybrids

Attention mechanisms (feature, temporal, self-, and multi-head attention) assign greater weight to selected features or time steps and are useful because attacks may be driven by a small subset of features or a short activity interval. However, attention weights should not automatically be presented as faithful explanations; attention can serve as a learned weighting mechanism, while SHAP or Integrated Gradients independently assess feature contribution. A 2024 self-attention 1D-CNN–LSTM study explicitly combines CNN, LSTM, self-attention, and SHAP-based feature analysis, confirming that this combination is already established in the literature.

5.6 Autoencoders and Variational Autoencoders

Autoencoders learn to reconstruct input data and flag anomalies via reconstruction error. N-BalIoT is the foundational IoT example, using deep autoencoders to detect Mirai and BASHLITE behaviour on nine infected commercial devices. Autoencoders can operate with benign-only training and are useful for zero-day research, but threshold selection is difficult, some attacks reconstruct well (evading detection), and reconstruction error is not inherently explainable. A practical hybrid pattern is autoencoder → latent representation → classifier.

5.7 Transformers

Transformers model relationships through self-attention rather than recurrence, offering strong long-range dependency modelling and parallel training, at the cost of memory, computational demand, and data-hungriness that are often unnecessary for modest tabular IoT datasets. A 2024 study on CIIoT2023 found Transformer performance particularly competitive for multiclass classification, although CNN–LSTM and DNN models performed better on some binary tasks — evidence that no single family dominates across all evaluation settings.

5.8 Graph Neural Networks

IoT traffic can be represented as a graph with devices/IPs/gateways as nodes and communications as edges. E-GraphSAGE was an early influential approach explicitly modelling edge features and topology for IoT NIDS, and recent spatial-temporal GNN work combines graph structure with sequence modelling and autoencoder pretraining, including healthcare-IoT applications. GNNs better represent device-to-device relationships and lateral movement than purely tabular models, but require reliable graph construction, more complex preprocessing, and substantially higher implementation cost, making them less suitable as a first undergraduate architecture.

6. Hybrid Deep Learning: Taxonomy and Literature

6.1 Taxonomy of 'Hybrid' in IoT IDS Research

Hybrid category	Examples	Purpose
Architecture-level	CNN–LSTM, CNN–GRU, CNN–BiLSTM, CNN–attention, CNN–Transformer	Combine local feature extraction with temporal/global modeling
Representation-level	Autoencoder–classifier, AE–CNN, AE–LSTM	Learn compact/denoised representations before classification
Algorithmic	ML + DL, ensemble + DL, supervised + unsupervised	Combine complementary decision mechanisms
Deployment-level	Edge + cloud, fog + cloud	Balance latency and computational capacity
Privacy-level	Federated + centralized	Preserve data locality while sharing learned information
Data-level	Network flows + logs + telemetry	Improve context and reduce blind spots
Optimization-level	DL + feature selection, DL + SMOTE, DL + pruning/quantization	Improve efficiency or minority-class performance

6.2 Representative Hybrid Deep Learning Literature

Table 6.1 summarizes representative hybrid-architecture papers directly relevant to the proposed project. Full bibliographic details appear in the References section.

Table 6.1 — Representative Hybrid Deep Learning + XAI Studies

Study (first author, year)	Architecture	Dataset	XAI	Critical limitation
Meidan et al., 2018	Deep autoencoder	N-BalIoT	None	Narrow attack families; small laboratory device population
Ferrag et al., 2022	Multiple ML/DL benchmark	Edge-IIoTset	None	Synthetic/testbed scenario and possible feature shortcuts
Sharma et al., 2024	DNN/CNN	IoT/NIDS datasets	SHAP, LIME	Explanation evaluation limited relative to detection evaluation
Sasi et al., 2024	1D-CNN–LSTM–self-attention	Multiple IoT datasets	SHAP	Very high results require leakage/split scrutiny

Study (first author, year)	Architecture	Dataset	XAI	Critical limitation
Ebrahimi et al., 2025	CNN	IoT traffic	SHAP	CNN lacks strong temporal justification
Fattah et al., 2025	LSTM–CNN + SMOTE	IoT IDS dataset	SHAP	Strong benchmark claims need leakage-aware validation
Arif et al., 2025	CNN–LSTM	SDN/IoT healthcare	SHAP	Narrow, domain-specific attack setting
Fatema et al., 2025	ANN + FedAvg	CICIoT2023	SHAP	Reported accuracy (88.2%) below centralized benchmarks
Dataset-centric FL study, 2026	LSTM/Transformer + FedAvg/FedProx/FedNova	Edge-IIoTset, CICIoT2023, TII-SSRC-23	None	Shows single-dataset FL results are insufficient for generalization claims
ST-GNN + AE pretraining, 2026	Autoencoder + spatial-temporal GNN	Healthcare IoT	Not central	High complexity for edge/undergraduate implementation

This body of evidence confirms that CNN–LSTM and CNN–GRU architectures are not, by themselves, novel; attention-enhanced hybrids, SHAP-based feature analysis, and combinations of hybrid architectures with XAI are already well represented in the 2018–2026 literature. Section 18 (Novelty Analysis) develops the implications of this observation for project design.

7. Explainable AI for Intrusion Detection

7.1 XAI Methods

Method	Local/global	Model-specific?	Main advantage	Main limitation
SHAP	Local and global	Model-agnostic (KernelSHAP); optimized variants exist	Strong theoretical foundation; additive contributions	Expensive for deep models; correlated features complicate interpretation
LIME	Local	Model-agnostic	Simple, intuitive, works with black boxes	Sampling instability; sensitive to neighborhood definition
Integrated Gradients	Local, aggregatable globally	Differentiable models	Computationally efficient; principled baseline path	Requires baseline choice; difficult with correlated tabular features
DeepLIFT	Local	Neural networks	Handles some saturation issues	Reference dependence; model-specific
Permutation importance	Global	Model-agnostic	Easy to explain	Correlated features and computational cost
Attention weights	Local/temporal	Model-specific	Naturally available in attention models	Not guaranteed to be faithful explanations
Counterfactuals	Local	Model-agnostic or model-specific	Answers 'what would need to change?'	May be physically or operationally meaningless

7.2 SHAP

SHAP assigns feature contributions based on Shapley values, approximating a prediction as the sum of a base value and per-feature contributions. SHAP can answer which features contributed to a specific alert, which features generally distinguish attack traffic, whether a feature increased or decreased attack probability, and whether the model relies on plausible security indicators. It is the most common XAI approach in recent IoT IDS work, often combined with LIME; a 2026 systematic review reports that SHAP and LIME dominate the XAI-for-IDS literature while identifying a persistent imbalance between detection performance, computational efficiency, and explanation evaluation.

7.3 LIME

LIME generates perturbed samples around an input and fits an interpretable local surrogate model. It is useful for explaining individual alerts but can vary across repeated runs because its neighbourhood is stochastically sampled. A 2024 study evaluating state-of-the-art IoT NIDS models found that explanations from different XAI methods often conflict and that models can be vulnerable to inductive bias arising from dataset artifacts.

7.4 Why 'SHAP = Explainability' Is Insufficient

Producing a SHAP plot demonstrates only that an attribution method was executed; it does not by itself establish that the attribution is faithful to the model, stable under small perturbations, causally meaningful, useful to analysts, valid under distribution shift, robust to adversarial manipulation, or computationally deployable. A 2026 study comparing SHAP and LIME under DeepFool perturbations found that SHAP generally produced stronger feature alignment and completeness, while LIME sometimes achieved better ranking stability but also generated more spurious attributions — direct evidence that explanation quality must be measured, not assumed.

8. Evaluating Explanation Quality

8.1 Prediction Performance versus Explanation Quality

Prediction performance (precision, recall, F1, false-positive/negative rate, macro-F1, PR-AUC, detection latency) and explanation quality (fidelity, faithfulness, stability, consistency, completeness, sparsity, robustness, analyst usefulness, explanation latency) are distinct dimensions. A detector can achieve high F1 while producing poor explanations, and conversely an interpretable model can offer only moderate predictive performance — a distinction central to this report's argument that explainability claims require dedicated evaluation.

8.2 Suggested Explanation Tests

- Fidelity test — remove/mask features identified as important and check whether the prediction changes.
- Stability test — repeat explanations under small input perturbations, different random seeds, and equivalent records; measure rank correlation or top-k overlap.

- Completeness test — check whether the sum of attribution values approximates the gap between prediction and baseline.
- Class-consistency test — compare explanations across multiple instances of the same attack class.
- Cross-model consistency — compare explanations from the proposed model and a strong baseline; large disagreement may reveal model-specific shortcuts.
- Analyst usefulness — a small human study assessing whether an analyst can verify an alert, identify the attack family, and distinguish false positives from true attacks.

A Minor Project can realistically implement fidelity and stability testing, while formal analyst studies are more appropriately deferred to future work.

9. IoT Intrusion Detection Datasets

Reported row counts vary by release, preprocessing pipeline, and whether packet-, flow-, or sampled-file versions are used, and should therefore be treated as approximate. Table 9.1 compares the datasets most relevant to this project.

Table 9.1 — Comparative Summary of IoT Intrusion Detection Datasets

Dataset (year)	IoT-specific?	Approx. scale	Attack coverage	Main strength	Main weakness
NSL-KDD (2009)	No	~148k train/test records	DoS, probe, R2L, U2R	Removes major KDD redundancy	Outdated threat model; not IoT
UNSW-NB15 (2015)	No (general NIDS)	2.5M raw records	9 families incl. DoS, exploits, reconnaissance	Modern relative to KDD; well documented	Not IoT-specific; scenario/feature shortcuts
N-BalIoT (2018)	Yes	~7M records, 115 features	Mirai, BASHLITE (10 attack types)	Authentic botnet behaviour from real infected devices	Narrow attack coverage; small device population
Bot-IoT (2019)	Yes	Very large (commonly sampled)	DDoS, DoS, reconnaissance, keylogging, exfiltration	Broad botnet attack categories	Extreme class imbalance
TON_IoT (2020)	Yes/IIoT	Multiple heterogeneous sets	9 broad attack categories	Multimodal (network + telemetry + OS)	Heterogeneous modalities complicate fair comparison
IoT-23 (2020)	Yes	>760M packets / 325M labeled flows	20 malware captures + 3 benign	Real malware captures, large scale	Capture-level and scenario bias
IoTID20 (2020)	Yes	~625k instances, ~80–86 features	DoS, Mirai, MITM ARP spoofing, scanning	Manageable size; hierarchical labels	Very small device diversity
Edge-IIoTset (2022)	Yes/IIoT	Large flow/log dataset; 61 selected features	14 attacks in 5 categories	Supports centralized and federated research	Feature/scenario leakage risk; testbed bias
CICIoT2023 (2023)	Yes	46,686,579 records, 47 features	33 attacks in 7 categories, 105 devices	Scale, diversity, contemporary IoT coverage	Severe imbalance; scenario shortcuts; expensive processing

Dataset (year)	IoT-specific?	Approx. scale	Attack coverage	Main strength	Main weakness
CICIoMT2024 (2024)	Yes (medical IoT)	Large; release-specific	18 attacks (DoS/DDoS, reconnaissance, MQTT, spoofing)	Healthcare-specific, multi-protocol testbed	Domain-specific; not general-purpose IoT

9.1 Dataset Recommendation for This Project

Edge-IIoTset is recommended as the primary dataset because it offers a manageable subset, IoT/IIoT context, multiple attack categories, device and protocol heterogeneity, and a realistic feature-selection exercise, with a natural extension to federated learning if time permits. CICIoT2023 is recommended as an optional large-scale or cross-dataset test, given its modern and broad attack coverage, provided that careful sampling, deduplication, and temporal/device-aware validation are applied. N-BaIoT is a suitable, smaller, and well-defined alternative for autoencoder-centred anomaly research, while IoTID20 is useful chiefly for early prototyping given its limited device and attack diversity. Datasets should not be combined blindly: differences in feature definitions, label semantics, protocols, and collection conditions can create artificial results, so any cross-dataset test should use only a small set of semantically compatible features.

10. Dataset Problems, Leakage, and Evaluation Protocols

10.1 Random Train/Test Split Risk

A random row-level split can place near-identical flows, adjacent time windows, or records from the same attack campaign into both training and test partitions, allowing a model to recognize device identity, source/destination IP, port, timestamp artifacts, or attack-generator behaviour rather than genuine attack semantics. The result is optimistic performance that does not reflect real-world generalization.

10.2 Common Leakage Sources

- Source/destination IP addresses acting as a proxy for the label.
- Destination port, when an attack targets only one service.
- Device identity learned from dataset-specific device labels.
- Timestamp, capture, or scenario identifiers that reveal labels.
- Duplicate or near-duplicate flows distributed across splits (memorization mistaken for detection).
- Feature scaling, selection, or SMOTE applied before splitting, letting test information leak into training.
- Attack-family leakage when multiple variants of the same campaign appear in every partition.

10.3 Evaluation Protocols, by Increasing Realism

Protocol	Realism	Use
Random stratified split	Low to moderate	Initial reproducibility baseline only
Group split by capture/session/device	Better	Prevents scenario and device memorization

Protocol	Realism	Use
Chronological split	High for deployment	Trains on earlier traffic, tests on later traffic
Cross-scenario validation	High	Tests transfer between environments
Cross-device validation	High	Tests device generalization
Cross-dataset validation	Very high but difficult	Tests semantic and domain transfer
Unseen-attack validation	High for anomaly research	Tests zero-day/family-level generalization

Recent dataset-centric work emphasizes chronological, cross-device, cross-scenario, and cross-dataset validation. A 2026 cross-domain evaluation found that performance can fall substantially under temporal and domain shift, with reported macro-F1 degradation of approximately 0.20–0.27 in some transfer settings, and a 2026 federated-learning evaluation argues that single-dataset FL results do not establish generalization, evaluating Edge-IIoTset, CICIOT2023, and TII-SSRC-23 to expose environmental differences. These findings directly motivate the multi-protocol evaluation design adopted in Section 20.

11. Class Imbalance

IoT IDS datasets are typically dominated by normal traffic and high-volume DDoS floods, with only a small number of rare but operationally serious attacks such as MITM, backdoor, or data-theft. Accuracy is misleading in this setting because a model can perform well on the majority class while missing minority attacks entirely.

Technique	Advantage	Risk
Class weighting	Simple; preserves all examples	May not resolve extreme class overlap
Random undersampling	Faster, balances classes	Discards information
SMOTE	Creates synthetic minority points	Can create unrealistic network records
ADASYN	Focuses on difficult minority regions	Can amplify noise
Focal loss	Emphasizes difficult examples	Requires careful tuning
GAN-based generation	Can model complex minority distributions	Synthetic artifacts and leakage risk

For this project, class weighting is recommended as the primary technique, supported by a controlled SMOTE comparison, with macro-F1 and per-class recall as reporting metrics, and with the strict rule that all resampling is fit on the training partition only, never on validation or test data.

12. Feature Engineering and Selection

Common IoT flow features include flow duration, forward/backward packet and byte counts, packet-length statistics, flow packets/bytes per second, inter-arrival time, TCP flag counts, protocol, ports, header lengths, active/idle times, and application-layer metadata (DNS, HTTP, MQTT). Feature-selection methods commonly used in the literature include correlation filtering, mutual information, chi-square, recursive feature elimination, tree-based importance, permutation importance, SHAP-based selection, and PCA,

each with characteristic strengths and cautions (for example, impurity-based tree importance can be biased, and PCA reduces direct interpretability).

A methodologically strong design constructs two feature sets: an operational feature set that excludes source/destination IP, capture ID, raw timestamp, explicit attack identifiers, and any field unavailable at deployment time; and a shortcut-inclusive feature set retained only for a diagnostic experiment demonstrating how leakage inflates results. The comparison between these two sets is itself a meaningful project contribution and is incorporated into the ablation plan in Section 21.

13. Deployment, Efficiency, and Broader System Considerations

13.1 Edge, Fog, and Cloud Trade-offs

Cloud deployment offers high compute and central visibility at the cost of latency, bandwidth, and privacy; edge deployment offers low latency and better data locality but is constrained in CPU, memory, and storage; fog/hybrid deployment sits between the two. A realistic student architecture is: IoT traffic → edge/gateway feature extraction → compact model → local explanation → dashboard. Physical IoT hardware is not required — a replay engine can stream flow records to a local inference service while a resource study measures model size, memory, latency, and throughput.

13.2 Lightweight Deep Learning

Relevant compression techniques include feature reduction, smaller CNN/GRU layers, depthwise-separable convolutions, structured/unstructured pruning, INT8 quantization, knowledge distillation, and early-exit inference. A 2026 edge-oriented study describes combining INT8 quantization, structured pruning, and knowledge distillation for lightweight anomaly detection. For this project, the practical extension is to train a full CNN–GRU, apply feature selection, compare FP32 versus post-training INT8 quantization, and report model size, latency, CPU, and memory — without claiming production-grade TinyML deployment unless tested on genuine constrained hardware.

13.3 Federated Learning

In federated learning, a central server initializes a model, IoT gateways train locally, clients transmit model updates rather than raw traffic, and the server aggregates updates (FedAvg being the standard baseline). Benefits include data locality and multi-organization collaboration; problems include non-IID client data, client drift, communication cost, and vulnerability to poisoning. FedProx (a proximal term reducing client drift) and FedNova (update normalization) are reported to improve stability and reduce communication rounds relative to FedAvg under heterogeneous clients. Recent work explicitly combines federated learning with SHAP by computing explanations locally and aggregating explanation statistics without exposing raw client data (e.g., FEX-IDS and related FedXAI-style systems), confirming that federated XAI is an active but non-trivial research direction requiring multi-client simulation, non-IID partitioning, and explanation-aggregation validation — better suited to future work than to the core Minor Project scope.

13.4 Concept Drift and Continual Learning

Concept drift occurs when the joint distribution of traffic and labels changes over time due to firmware updates, new device models, seasonal traffic, protocol changes, or attacker adaptation. Recent research explores incremental learning, drift detection, memory replay, and adaptive ensembles. A feasible Minor Project experiment trains on an early time window, tests on a later window, measures degradation, and optionally fine-tunes on a small later window to assess whether adaptation improves macro-F1 without catastrophic forgetting.

13.5 Adversarial Machine Learning

AI-based IDS are themselves attackable through evasion (modifying traffic features while preserving attack functionality), poisoning (inserting malicious training samples, particularly dangerous in online/federated/continual settings), model extraction, and explanation attacks (using SHAP/LIME/attribution outputs to infer which features or thresholds to avoid). Recent work explicitly studies adversarial manipulation of SHAP, LIME, and Integrated Gradients explanations in cybersecurity contexts, and a 2026 evaluation of SHAP and LIME under DeepFool perturbations emphasizes that high detection accuracy does not guarantee high-fidelity explanations. For this project, adversarial robustness is best treated as future work or, at most, a small perturbation-based robustness experiment.

14. Evaluation Metrics

For binary detection, standard confusion-matrix quantities (true/false positive/negative) yield accuracy, precision, recall, F1, false-positive rate (FPR), and false-negative rate (FNR). Because accuracy is misleading under severe class imbalance, macro-F1 (unweighted mean of per-class F1, essential for rare attacks), weighted-F1, ROC-AUC, and PR-AUC (more informative when the positive class is rare) should also be reported.

A credible report should present accuracy alongside precision, recall, F1, macro-F1, weighted-F1, per-class precision/recall/F1, FPR, FNR, the confusion matrix, ROC-AUC/PR-AUC where appropriate, and confidence intervals or repeated-seed variation — together with deployment-oriented measures (training time, inference latency, throughput, model size, peak memory, CPU/GPU requirement, and explanation latency). A 2025 IoT security-audit framework illustrates the importance of combining detection results with resource measures, showing that high overall multiclass performance can coexist with weak performance on serious minority attacks such as MITM — a pattern the proposed methodology is explicitly designed to expose rather than obscure.

15. Literature Review Matrix

Table 15.1 consolidates the most directly relevant studies identified during this review. It should be expanded during Stage-2 documentation through additional database searches and full-text verification; DOIs are provided where available to support independent verification.

Table 15.1 — Literature Review Matrix (Selected, 2018–2026)

No.	Study	Year	Dataset	Architecture / Hybrid	XAI	Key limitation
1	Meidan et al., N-BaloT	2018	N-BaloT	Deep AE	No	Narrow attack families; 9 devices only
2	Koroniotis et al., Bot-IoT	2019	Bot-IoT	Dataset paper	No	Severe imbalance; synth scenario dependence
3	Alsaedi et al., TON_IoT	2020	TON_IoT	Dataset (multimodal)	No	Complex modalities; sce bias
4	Ullah & Mahmoud, IoTID20	2020	IoTID20	Flow-based dataset	No	Limited devices and atta diversity
5	Garcia et al., IoT-23	2020	IoT-23	Malware captures	No	Capture and malware-fa bias
6	Ferrag et al., Edge-IIoTset	2022	Edge-IIoTset	ML/DL benchmark	No	Testbed and feature-lea risk
7	Sarhan et al., NF standard feature set	2022	NF datasets	NetFlow normalization (data-level hybrid)	No	Derived data inherits sou dataset limits
8	Alsoufi et al., SLR	2021	Multiple	Survey	Limited	Older evidence base; lim recent XAI coverage
9	Sharma et al., XAI-IDS	2024	IoT/NIDS	DNN/CNN	SHAP, LIME	Explanation evaluation v than detection evaluation
10	Sasi et al., self-attention CNN-LSTM	2024	Multiple IoT	1D-CNN-LSTM-attention	SHAP	Very high results need le aware replication
11	Ali & Zhang, XAI-enabled IDS	2024	CICIoT2023	DL framework	XAI	Limited explanation valid
12	Ebrahimi et al., explainable CNN	2025	IoT dataset	CNN	SHAP	CNN not inherently temp
13	Fattah et al., hybrid LSTM-CNN	2025	IoT IDS dataset	LSTM-CNN + SMOTE	SHAP	Benchmark generalization/deployme unclear
14	Arif et al., explainable CNN-LSTM DDoS	2025	SDN/IoT healthcare	CNN-LSTM	SHAP	Narrow application scop
15	Fatema et al., Federated XAI IDS	2025	CICIoT2023	ANN + FedAvg	SHAP	Reported accuracy (88.2 below centralized bench
16	FEX-IDS	2025	IoT	FL model	SHAP/LIME	Global explanation aggre remains challenging
17	IoT stream IDS with XAI	2025	TON_IoT	1D-CNN/DNN/TabNet	XAI	Near-100% results sugge for split scrutiny
18	Explainability evaluation of SOTA IoT NIDS	2024	IoT NIDS models	Multiple (comparison)	TRUSTEE/SHAP	Preprint; needs peer-rev confirmation
19	E-GraphSAGE	2021	Four NIDS datasets	GNN	No	Graph construction and deployment complexity
20	Transformer IDS on CICIoT2023	2024	CICIoT2023	Transformer	No	Strong accuracy but limi resource evaluation
21	Dataset-centric FL evaluation	2026	Edge-IIoTset, CICIoT2023, TII-SSRC-23	LSTM/Transformer + FL	No	Does not center XAI

No.	Study	Year	Dataset	Architecture / Hybrid	XAI	Key limitation
22	Cross-dataset temporal/semantic generalization	2026	Four datasets	RF, GB, MLP, AE, CNN	No	Broader NIDS scope, not only
23	DeepFool robustness of XAI in IoT IDS	2026	Bot-IoT, Edge-IIoT, N-BaIoT	CNN/DNN/LSTM/RF	SHAP/LIME	Adversarial setup may not reflect all operational attacks
24	IoT IDS systematic review	2026	Literature review	Survey	SHAP/LIME dominant	Recommendations require empirical validation
25	ST-GNN + AE healthcare IDS	2026	Healthcare IoT	AE + spatial-temporal GNN	Not central	High computational and complexity

16. State-of-the-Art Synthesis

State of the art should not be equated with the architecture reporting the highest accuracy on a single random split. Synthesizing across families:

- CNN-based models are efficient, strong on flow-level features, and a strong practical edge-deployment baseline, but have limited long-range temporal modelling.
- CNN-LSTM/CNN-GRU are common and effective but are saturated as a novelty claim and may be unjustified when data are not genuine sequences.
- Attention-based hybrids improve feature/time-step focus but should not be equated with faithful explanations.
- Transformers are strong for long sequences and large datasets but carry substantial resource cost, better suited to large than to small student datasets.
- GNNs align best with communication topology and are promising for lateral-movement detection but require sophisticated graph engineering.
- Autoencoders remain important for benign-only and unknown-attack detection, but reconstruction-threshold selection and false-positive control remain open problems.
- Federated IDS are important for privacy and multi-organization settings but are not automatically more accurate than centralized training and require rigorous non-IID and poisoning evaluation.
- Explainable IDS most commonly follow the pattern DL classifier + SHAP/LIME post-hoc explanation, which is useful but insufficient for a genuine state-of-the-art claim; a stronger explainable IDS additionally reports explanation fidelity, stability, latency, robustness, and cross-dataset behaviour alongside detection metrics.

17. Research Gaps

Ten gaps recur across the reviewed literature, each assessed here for undergraduate feasibility.

Gap 1 — High benchmark performance but weak generalization

Models often perform well under random splits but degrade under chronological, cross-device, or cross-dataset evaluation, which matters because production IDS encounter future traffic, new devices, and

evolving attacks. Feasibility: highly feasible. Proposed solution: compare random, chronological, and device/group splits, with an optional external-dataset test.

Gap 2 — Dataset leakage

IP addresses, ports, device identities, capture IDs, duplicates, and timestamps can act as label proxies, causing the model to memorize the test scenario instead of learning attack behaviour. Feasibility: highly feasible. Proposed solution: a feature-hygiene ablation (all features → remove identifiers → remove ports → remove timestamps → remove highly correlated features).

Gap 3 — XAI without explanation-quality evaluation

Most papers generate SHAP or LIME figures without evaluating faithfulness, stability, robustness, or usefulness, risking analyst trust in incomplete or misleading explanations. Feasibility: highly feasible. Proposed solution: feature-removal fidelity, repeated-seed stability, top-k overlap, and explanation-latency measurement.

Gap 4 — Accuracy-centric evaluation

A 99% accuracy figure can hide poor minority-class recall or excessive false alerts, which matters because rare attacks can be operationally severe. Feasibility: highly feasible. Proposed solution: report macro-F1, per-class recall, FPR, FNR, PR-AUC, and confusion matrices.

Gap 5 — Hybrid complexity without ablation

Papers often add CNN, LSTM, attention, and feature selection without demonstrating which component actually helps. Feasibility: highly feasible. Proposed solution: systematic ablation across CNN, GRU/LSTM, CNN-GRU, and CNN-GRU-attention, with and without feature selection and class weighting.

Gap 6 — Poor edge suitability

Many studies claim real-time or lightweight deployment without measuring memory, latency, throughput, or model size. Feasibility: feasible. Proposed solution: measure CPU inference latency, throughput, model size, peak RAM, and optional quantized inference.

Gap 7 — Lack of unseen-attack evaluation

Supervised models generally recognize only attack families seen during training. Feasibility: moderately feasible. Proposed solution: hold out one attack subtype during training as an exploratory generalization test, without overstating it as a complete zero-day solution.

Gap 8 — Concept drift

Traffic and attacks evolve over time. Feasibility: moderate. Proposed solution: chronological evaluation plus one limited incremental-update experiment.

Gap 9 — Adversarial robustness

Attackers can manipulate flow features or explanations. Feasibility: limited but possible as future work. Proposed solution: controlled perturbations of packet rate, duration, and packet length, without claiming comprehensive adversarial defence.

Gap 10 — Federated XAI privacy trade-offs

Local explanations, global aggregation, and privacy preservation can conflict. Feasibility: low for a core Minor Project. Proposed solution: treat as future work or, at most, a small simulation.

18. Hybrid Deep Learning + XAI Novelty Analysis

18.1 Already Common (Not a Strong Novelty Claim)

- CNN–LSTM or CNN–GRU for IoT IDS.
- Adding attention to CNN–LSTM.
- Applying SHAP or LIME to a trained IDS and reporting feature importance.
- Using SMOTE with a DL classifier.
- Evaluating on CICIoT2023 alone.
- Claiming 'real-time' operation without system measurements.

18.2 Saturated Combinations

- CNN + LSTM + SHAP; CNN + LSTM + attention; CNN + GRU + SMOTE.
- LSTM + SHAP; CNN + SHAP; deep learning + LIME + SHAP.
- Federated learning + SHAP; CICIoT2023 + deep classifier + SHAP.

18.3 Less Explored and More Defensible

- Leakage-aware evaluation of a hybrid model.
- Explanation fidelity and stability evaluation.
- Chronological and device-group validation.
- Explicit comparison of shortcut-prone versus operational feature sets.
- Detection–efficiency–explainability trade-off analysis.
- Cross-dataset testing with a standardized common feature subset.
- Minority-attack-focused analysis rather than overall accuracy optimization.

18.4 What Would Not Count as Meaningful Novelty

- Replacing LSTM with GRU without a research question.
- Adding a dense layer or changing kernel size without ablation.
- Applying SHAP after training and calling the resulting system novel.
- Reporting a marginally higher accuracy on a random split.
- Claiming 'first explainable IoT IDS' without a systematic literature search.

18.5 Recommended Contribution Statement

'This project develops and evaluates a compact hybrid deep-learning IDS for IoT flow data, with explicit feature-hygiene controls, class-imbalance handling, SHAP-based global and local explanations, and comparison between random and chronological validation. The contribution is a reproducible assessment of the trade-off between detection performance, generalization, explanation quality, and computational cost.' This framing is academically meaningful without overstating architectural novelty, and is consistent with how the field's genuine gaps (Section 17) were characterized.

19. Proposed Project Architecture and Recommended Direction

19.1 Recommended Direction

Leakage-Aware and Explainable CNN–GRU Intrusion Detection for IoT Networks: A Comparative Study under Random and Chronological Validation.

This direction is recommended because it is directly aligned with the project's title, implementable using public datasets, appropriately scoped for undergraduate development, more defensible than inventing another complex architecture, suitable for both binary and multiclass detection, compatible with SHAP, and easily demonstrated through a dashboard and lightweight-deployment measurements.

CNN–GRU is preferred over a larger model because GRU generally uses fewer parameters than LSTM while retaining gated temporal modelling, and CNN provides local feature extraction; the resulting architecture is easier to train and deploy than a Transformer or GNN. The project explicitly does not claim CNN–GRU as a novel architecture — its contribution lies in the evaluation design: leakage-aware features, chronological validation, macro-F1 and minority-class analysis, explanation fidelity/stability, efficiency measurement, and ablation studies.

19.2 Dataset Plan

- Primary dataset: Edge-IIoTset.
- Optional external/cross-dataset test: CICIoT2023, restricted to semantically compatible flow features.
- Fallback for limited storage/compute: IoTID20 for prototyping, followed by a smaller Edge-IIoTset validation subset.

19.3 System Architecture

The proposed pipeline is: IoT network flows → flow feature extraction → feature hygiene and schema validation → removal of identifiers and leakage-prone fields → missing-value handling and categorical encoding → feature selection → class-weighted training → 1D-CNN feature extractor → GRU temporal encoder → dense classification head → binary/multiclass prediction → SHAP global and local explanation → explanation quality tests → dashboard and alert record.

Preprocessing

Remove labels/metadata and IP addresses, timestamps, capture identifiers, and direct device identifiers; encode protocol and categorical fields; impute missing values; standardize continuous variables using training-set statistics only; and persist all preprocessing objects for reproducibility.

Feature selection

Combine correlation filtering, mutual-information or tree-based importance, and SHAP-based ranking from a baseline model to produce a single fixed selected-feature subset used consistently across all experiments.

Model configuration (starting point, not a claimed optimum)

- Input: selected flow-feature sequence.
- Conv1D: 32 filters, kernel size 3; batch normalization; ReLU; max pooling; dropout 0.2.
- GRU: 64 units.
- Dense: 32 units, ReLU; dropout 0.2.
- Output: sigmoid (binary) or softmax (multiclass).
- Optimizer: Adam, initial learning rate 1e-3 (tuned on validation data); batch size 128–256; early stopping and learning-rate reduction on validation macro-F1; fixed seeds with repeated runs.

Explainability module

SHAP global summary and class-specific importance; local waterfall plots for true positives, false positives, false negatives, and minority attacks; feature-removal fidelity; repeated-run stability; and explanation latency. Optional extensions: LIME comparison on a small test subset, or Integrated Gradients for a model-specific method comparison.

20. Experimental Methodology

20.1 Data Preparation

- Download the official dataset release and record version, filenames, hashes, and source URL.
- Inspect labels, missing values, duplicates, and class counts.
- Remove unusable or direct-label fields; define the operational feature policy before training.
- Group flows by capture/scenario/device/time where metadata allows.
- Split the data before fitting scalers, feature selectors, or resampling methods.

20.2 Data Splits

- Baseline: stratified random split — 70% train / 15% validation / 15% test.
- Main realistic split: chronological — earliest 60% train / next 20% validation / latest 20% test.
- Group split: by capture/session/device where possible.
- Optional cross-dataset test: train on Edge-IIoTset, evaluate on a compatible CICIOT2023 feature subset (measuring domain shift, not expecting high performance).

20.3 Baseline Models

Mandatory baselines: Logistic Regression, Decision Tree, Random Forest, XGBoost (if computationally practical), MLP, 1D-CNN, GRU or LSTM, and the proposed CNN-GRU.

20.4 Experiment Matrix

Table 20.1 — Planned Experiment Matrix

ID	Model / condition	Purpose
E1	Logistic Regression	Linear baseline
E2	Random Forest	Strong tabular baseline
E3	XGBoost	High-performance tree baseline
E4	MLP	Dense DL baseline
E5	1D-CNN	Convolution-only model
E6	GRU	Temporal-only model
E7	CNN-GRU	Proposed hybrid
E8	CNN-GRU + class weights	Imbalance-handling effect
E9	CNN-GRU + SMOTE	Resampling comparison
E10	CNN-GRU + feature selection	Efficiency and generalization
E11	CNN-GRU, random split	Conventional benchmark
E12	CNN-GRU, chronological split	Temporal robustness
E13	CNN-GRU, group split	Scenario/device generalization
E14	CNN-GRU + SHAP	Explainability
E15	CNN-GRU + fidelity/stability tests	Explanation quality
E16	Quantized CNN-GRU (subset)	Edge-efficiency study

20.5 Metrics and Statistical Reporting

Detection metrics: accuracy, precision, recall, F1, macro-F1, weighted-F1, per-class recall, FPR, FNR, PR-AUC, ROC-AUC, confusion matrix. Deployment metrics: training time, inference latency (per flow and batch), throughput, parameter count, serialized model size, peak memory, CPU utilization, explanation latency, and quantized-versus-unquantized comparison. Each experiment should be run with at least three random seeds, reporting mean, standard deviation, and, where feasible, confidence intervals — never only the single best run.

20.6 Ablation Study

The most important ablations are: (1) random versus chronological split; (2) leakage-prone versus operational features; (3) CNN versus GRU versus CNN-GRU; (4) with versus without class weighting; and (5) SHAP fidelity and stability. The broader ablation set additionally covers CNN+GRU+attention, SMOTE, full versus reduced sequence length, and FP32 versus INT8 precision, each mapped to the specific claim it is designed to support.

20.7 Implementation Roadmap

- Phase 1 — Research: finalize the research question, complete the literature matrix, define the project's operational meaning of 'hybrid', and fix the dataset/feature/evaluation policy before implementation.
- Phase 2 — Dataset preparation: obtain official files, verify schema and labels, profile class distribution, detect duplicates, remove leakage-prone columns, build a reproducible preprocessing pipeline.
- Phase 3 — Baseline models: Logistic Regression, Random Forest, XGBoost, MLP, 1D-CNN, GRU/LSTM.
- Phase 4 — Proposed hybrid model: implement CNN–GRU, tune only on validation data, track parameters/training time/memory, run repeated-seed experiments.
- Phase 5 — XAI: implement SHAP, produce global importance, analyze representative correct and misclassified samples, compare majority- versus minority-attack explanations.
- Phase 6 — Evaluation: random, chronological, and group/device splits across binary and multiclass tasks and imbalance strategies.
- Phase 7 — Error analysis: false positives/negatives, rare classes, confusion between related attacks, explanation disagreement, dataset-shift failures.
- Phase 8 — Documentation and demonstration: a dashboard showing prediction, confidence, attack class, top features, and explanation, with explicit statement of the system's research-prototype status and honest documentation of limitations and non-novel aspects.

21. Generalization Testing

Generalization is treated as a first-class evaluation axis rather than an afterthought. Three complementary tests are proposed: (i) temporal generalization, via the chronological split described in Section 20.2, which measures degradation as the model encounters traffic generated after its training window; (ii) identity/scenario generalization, via group splitting by capture, session, or device, which measures whether the model has learned attack behaviour rather than dataset-specific identifiers; and (iii) domain generalization, via an optional cross-dataset test training on Edge-IIoTset and evaluating on a semantically compatible CICIOT2023 feature subset, which measures transfer across independently collected environments. Consistent with the literature reviewed in Section 10.3, performance is expected to fall under each of these tests relative to a random split, and the magnitude of that fall — not the random-split number alone — is treated as the primary evidence of the model's real-world reliability.

22. Limitations

- Public datasets are not equivalent to live IoT deployments.
- Flow-based features may omit payload semantics relevant to injection- or content-based attacks.
- A laboratory testbed cannot reproduce every real-world environment.
- SHAP explains the trained model's behaviour, not necessarily the true underlying cause of an attack.
- Chronological validation within one dataset does not guarantee cross-network generalization.

- A CNN–GRU architecture may be inappropriate when the data are not genuinely sequential.
- Class weighting and SMOTE can alter the effective class prior.
- A demonstration dashboard does not by itself make the IDS operationally trustworthy.
- Edge measurements on a laptop are not equivalent to measurements on every IoT gateway class.
- Attack labels in public datasets may be incomplete or scenario-dependent.

23. Future Scope

- Cross-dataset semantic feature alignment.
- Unseen-attack and open-set detection.
- Continual learning under concept drift.
- Federated learning with non-IID clients, and federated XAI.
- Adversarial evasion and poisoning robustness, and secure explanation generation.
- Graph-based topology modelling (GNN/temporal-GNN).
- Multimodal fusion of flows, logs, telemetry, and device state.
- Quantization-aware training and Raspberry Pi/TinyML deployment.
- Human analyst studies and explanation calibration/causal explanations.
- Evaluation using newer IoT and IoMT datasets as they are released.

24. Conclusion

The field of IoT intrusion detection is no longer primarily seeking another architecture that achieves 99% accuracy on a random split. The central open problem is the gap between benchmark performance and trustworthy operational detection, a gap this report frames as a function of detection quality, generalization, efficiency, explanation quality, and robustness considered jointly rather than in isolation. For a university Minor Project, the most valuable and realistic contribution is not a new collection of neural layers but a carefully controlled study that shows how a hybrid model performs against strong traditional and deep-learning baselines; how results change after removing leakage-prone features; how performance changes under chronological or group-based testing; whether imbalance handling improves minority-attack detection; whether SHAP explanations are faithful and stable; what computational cost the model and its explanation module introduce; and which claims remain valid outside the original random-split benchmark. This scope — a leakage-aware, explainable CNN–GRU IoT IDS evaluated under random, chronological, and group-based protocols — is technically achievable within a Minor Project timeline, academically defensible against the literature surveyed in Sections 1–18, and directly aligned with the genuine gaps identified in the 2018–August 2026 literature.

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